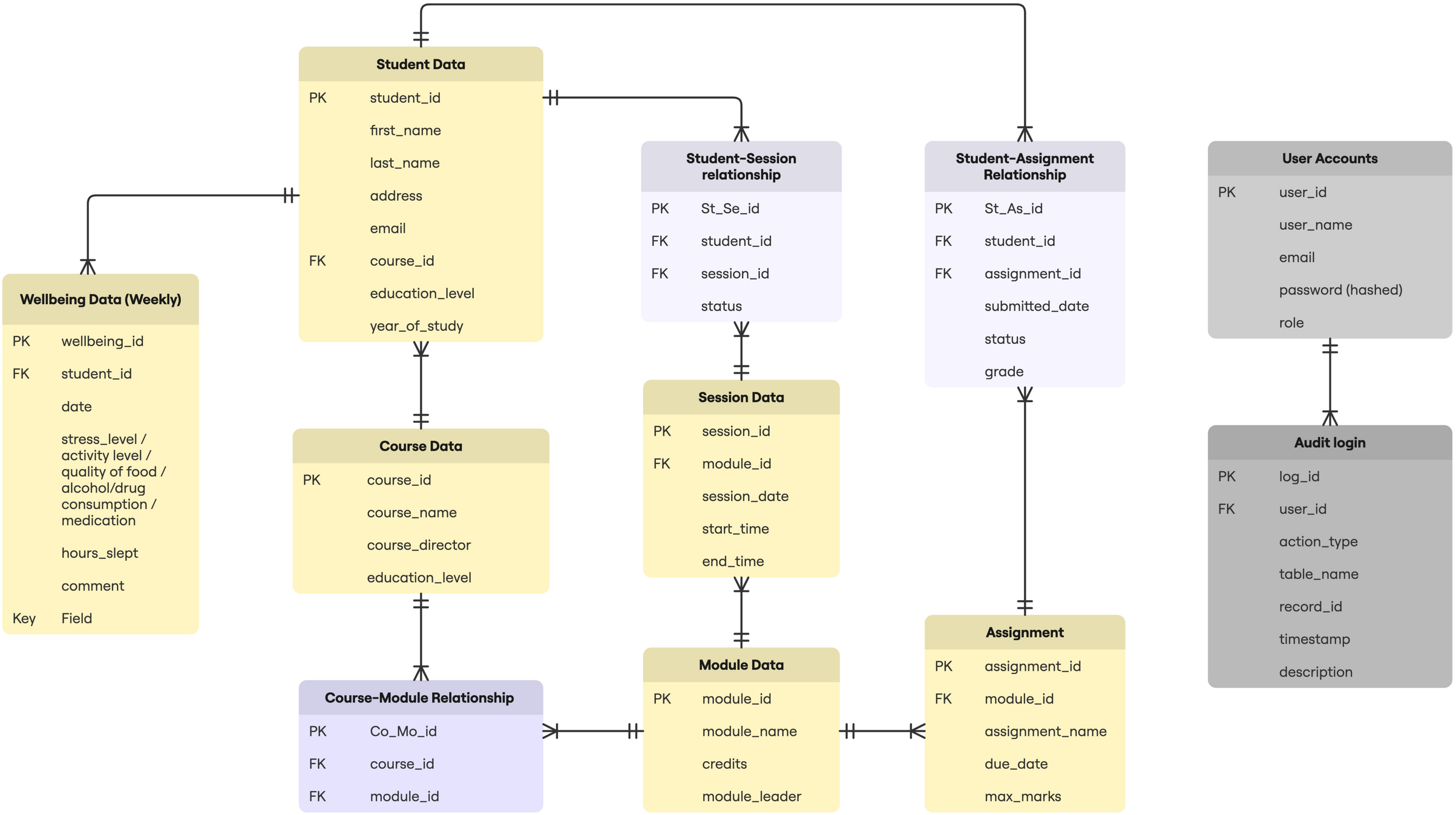
Entity Relationship Diagram



UML Use Case Diagram

Student Data System

Import data

Manage academic data

Manage wellbeing data

View individual performance

View individual wellbeing data

Log in and validate role

See early warning and alerts

View aggregated wellbeing analytics

View course analytics

Store data

Calculate engagement and wellbeing metrics

Staff/Admin

Student Wellbeing Officer

Course Director

<<Service>> University SSO

<<Service>> Reporting Export Service

Association

Include

Director Actions

Wellbeing actions

Staff/Admin actions

UML Class Diagram (overall)

Entities

Assignment

- assignment_id : int
- module_id : int
- name : str
- due_date : date
- max_marks : int
- students : List[Student]

+ __init__(assignment_id: int, module_id: int, name: str, due_date: date, max_marks: int)
+ add_student(student: Student) : None

AuditLog

- log_id : int
- user_id : int
- action_type : str
- table_name : str
- record_id : int
- timestamp : datetime

+ __init__(log_id: int, user_id: int, action_type: str, table_name: str, record_id: int, timestamp: datetime, description: str)

Course

- course_id : int
- name : str
- director : str
- education_level: str
- modules : List[Module]

+ __init__(course_id: int, name: str, director: str, education_level: str)
+ add_module(module: Module) : None

Module

- module_id : int
- name : str
- credits : int
- leader : str
- courses : List[Course]
- assignments : List[Assignment]

+ __init__(module_id: int, name: str, credits: int, leader: str)
+ add_course(course: Course) : None
+ add_assignment(assignment: Assignment) : None
+ add_session(session: Session) : None

Session

- session_id : int
- module_id : int
- start_time : time
- end_time : time
- date : date
- students : List[Student]

+ __init__(module_id: int, start_time: time, end_time: time, date: date)
+ add_student(student: Student) : None

Student

- student_id : int
- first_name : str
- last_name : str
- address : str
- course_id : int
- education_level : str

+ __init__(student_id: int, first_name: str, last_name: str, address: str, course_id: int, education_level: str, year_of_study: int, email: str)
+ add_session(session: Session) : None

UserAccount

- user_id : int
- username : str
- email : str
- role : str
- logs : List[AuditLog]

+ __init__(user_id: int, username: str, email: str, role: str)
+ add_log(log: AuditLog) : None

Wellbeing

- wellbeing_id : int
- student_id : int
- date : date
- stress : int
- activity : int
- food_quality : int

+ __init__(wellbeing_id: int, student_id: int, date: date, stress: int, activity: int, food_quality: int, alcohol_drugs: int, hours_slept: float, comment: str)

StudentSession

- student_id: int
- session_id: int
- status: str
- notes: str
- timestamps: List[datetime]

+ __init__(student_id: int, session_id: int, status: str)
+ mark_present()
+ mark_absent()
+ mark_late()
+ add_note(note: str)

StudentAssignment

- student_id: int
- assignment_id: int
- status: str
- grade: float
- submission_notes: str

+ __init__(student_id: int, assignment_id: int, status: str, grade: float = None)
+ mark_submitted()
+ mark_pending()
+ mark_late()
+ add_grade(grade: float)

Services

AssignmentServices

- repo: AssignmentRepo
+ __init__(db)
+ getBy_id(id: int) : Assignment | None
+ get_all() : List[Assignment]
+ get_module_assignments(module_id: int) : List[Assignment]

AuditLogServices

- repo: AuditLogRepo
+ __init__(db)
+ getBy_id(id: int) : AuditLogin | None
+ get_all() : List[AuditLogin]
+ get_user_logs(user_id: int) : List[AuditLogin]

CourseServices

- repo: CourseRepo
+ __init__(db)
+ getBy_id(id: int) : Course | None
+ get_all() : List[Course]
+ get_course_names() : List[str]

ModuleServices

- repo: ModuleRepo
+ __init__(db)
+ getBy_id(id: int) : Module | None
+ get_all() : List[Module]
+ get_course_names() : List[str]

SessionServices

- repo: SessionRepo
+ __init__(db)
+ getBy_id(id: int) : Session | None
+ get_all() : List[Session]
+ get_module_sessions(module_id: int) : List[Session]

StudentServices

- repo: StudentRepo
+ __init__(db)
+ getBy_id(id: int) : Student | None
+ get_all() : List[Student]
+ get_course_students(course_id: int) : List[Student]

UserAccountServices

- repo: UserAccountRepo
+ __init__(db)
+ getBy_id(id: int) : UserAccount | None
+ get_all() : List[UserAccount]
+ get_user_by_email(email: str) : UserAccount | None

WellbeingServices

- repo: WellbeingRepo
+ __init__(db)
+ getBy_id(id: int) : Wellbeing | None
+ get_all() : List[Wellbeing]
+ get_student_wellbeing(student_id: int) : List[Wellbeing]

StudentSessionServices

- repo: StudentsSessionRepo
+ __init__(db)
+ getBy_id(student_id: int, session_id: int) : StudentSession | None
+ get_all() : List[StudentSession]
+ get_student_sessions(student_id: int) : List[StudentSession]
+ mark_present(student_id: int, session_id: int)
+ mark_absent(student_id: int, session_id: int)
+ mark_late(student_id: int, session_id: int)

StudentAssignmentServices

- repo: StudentAssignmentRepo
+ __init__(db)
+ getBy_id(student_id: int, assignment_id: int) : StudentAssignment | None
+ get_all() : List[StudentAssignment]
+ get_student_assignments(student_id: int) : List[StudentAssignment]
+ mark_submitted(student_id: int, assignment_id: int)
+ mark_pending(student_id: int, assignment_id: int)
+ mark_late(student_id: int, assignment_id: int)
+ add_grade(student_id: int, assignment_id: int, grade: float)

Interface Repositories

<<interface>> IAssignmentRepo

(no attributes)

inherits all methods from IBaseRepo<Assignment>

<<interface>> IAuditLogRepo

(no attributes)

inherits all methods from IBaseRepo<AuditLog>

<<interface>> ICourseRepo

(no attributes)

inherits all methods from IBaseRepo<Course>

<<interface>> IModuleRepo

(no attributes)

inherits all methods from IBaseRepo<Module>

<<interface>> ISessionRepo

(no attributes)

inherits all methods from IBaseRepo<Session>

<<interface>> IStudentRepo

(no attributes)

inherits all methods from IBaseRepo<Student>

<<interface>> IBaseRepo<T>

(no attributes)

+ get_by_id(entity_id: int) : T | None
+ get_all() : List<T>
+ add(entity: T) : T
+ update(entity: T) : T

<<interface>> IUserAccountRepo

(no attributes)

inherits all methods from IBaseRepo<UserAccount>

<<interface>> IWellbeingRepo

(no attributes)

inherits all methods from IBaseRepo<Wellbeing>

<<interface>> IStudentSessionRepo

(no attributes)

inherits all methods from IBaseRepo<StudentSession>

<<interface>> IStudentASsignmentRepo

(no attributes)

inherits all methods from IBaseRepo<StudentAssignment>

Repositories

AssignmentRepo

Inherited:

- session : Session

+ AssignmentRepo(session: Session)

AuditLogRepo

Inherited:
- session : Session
- model : Type<AuditLog>

+AuditLogRepo(session: Session)

BaseRepo<T>

- session: Session
- model: Type<T>

+ __init__(session: Session, model: Type<T>)
+ get_by_id(id: int): T
+ get_all(): List<T>

CourseRepo

Inherited:
- session : Session
- model : Type<Course>

+CourseRepo(session: Session)

ModuleRepo

Inherited:
- session : Session
- model : Type<Module>

+ ModuleRepo(session: Session)

SessionRepo

Inherited:
- session : Session
- model : Type<Session>

+ <Session> (session: Session)
+ add(entity: Session) : Session

StudentRepo

Inherited:
- session : Session
- model : Type<Student>

+ StudentRepo(session: Session)

UserAccountRepo

Inherited:
- session : Session
- model : Type<UserAccount>

+ UserAccountRepo(session: Session)

WellbeingRepo

Inherited:
- session : Session
- model : Type<Wellbeing>

+ WellbeingRepo(session: Session)

StudentSessionRepo

Inherited:
- session : Session

+ get_by_student_and_session(student_id: int, session_id: int) : StudentSession | None
+ mark_attendance(entity: StudentSession) : StudentSession | None
+ get_attendance_for_student(student_id: int) : List[StudentSession]
+ get_attendance_for_session(session_id: int) : List[StudentSession]
+ get_attendance_report_for_module(module_id: int) : List[StudentSession]

StudentAssignmentRepo

Inherited:
- session : Session

+ get_by_student_and_assignment(student_id: int, assignment_id: int) : StudentAssignment | None
+ get_all_by_student(student_id: int) : List[StudentAssignment]
+ get_all_by_assignment(assignment_id: int) : List[StudentAssignment]
+ get_performance_report_for_module(module_id: int) : List[StudentAssignment]

DTOs

Tables

Assignment

- assignment_id : int (PK)
- module_id : int (FK)
- name : str
- due_date : date
- max_marks : int
+ module: Module
+ student_assignments: List<StudentAssignment>

AuditLog

- log_id: Integer (PK)
- user_id: Integer (FK)
- action_type: String
- table_name: String
- record_id: Integer
- timestamp: DateTime
- description: Text
+ user: UserAccount

Course

- course_id: Integer (PK)
- course_name: String
- course_director: String
- education_level: String
+ students: List<Student>
+ modules: List<Module>

Module

- module_id: Integer (PK)
- module_name: String
- credits: Integer
- module_leader: String
+ sessions: List<Session>
+ assignments: List<Assignment>
+ courses: List<Course>

Session

- session_id: Integer (PK)
- module_id: Integer (FK)
- session_date: Date
- start_time: Time
- end_time: Time
+ module: Module
+ student_sessions: List<StudentSession>

Student

- student_id: Integer (PK)
- first_name: String
- last_name: String
- address: Text
- email: String
- course_id: Integer (FK)
- education_level: String
- year_of_study: Integer
+ wellbeing: List<Wellbeing>
+ course: Course
+ student_sessions: List<StudentSession>

UserAccount

- user_id: Integer (PK)
- user_name: String
- email: String
- password: String
- role: String
+ audit_logs: List<AuditLogin>

Wellbeing

- wellbeing_id: Integer (PK)
- student_id: Integer (FK)
- date: Date
- stress_level: String
- activity_level: String
- quality_of_food: String
- alcohol_drug_consumption: String
- medication: String
- hours_slept: Integer
- comment: Text
+ student: Student

StudentSession

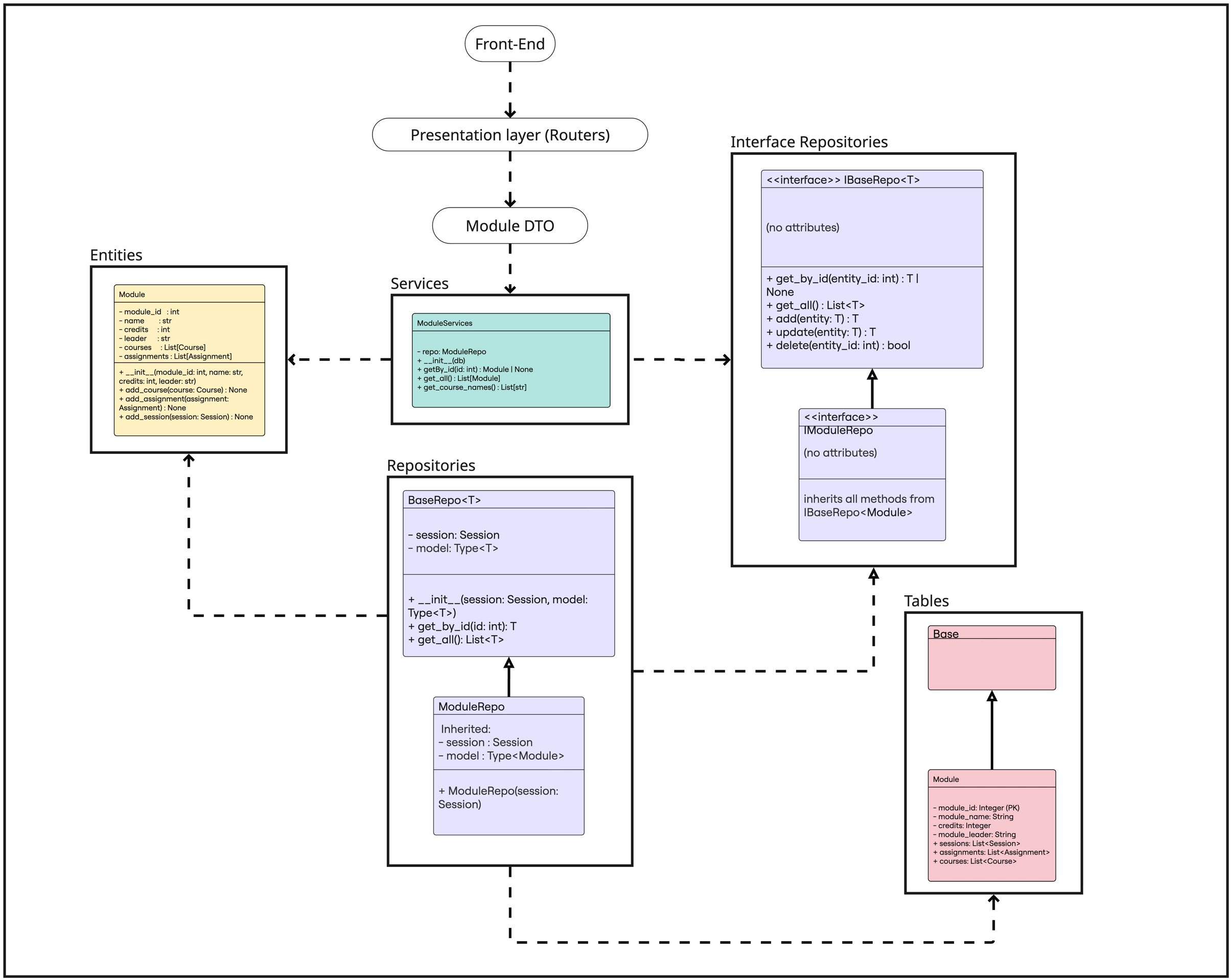
- st_se_id: int
- student_id: int
- session_id: int
- status: str

+ student: Student
+ session: Session

StudentAssignment

- st_as_id: int
- student_id: int
- assignment_id: int
- submitted_date: DateTime
- status: str
- grade: int

+ student: Student
+ session: Session

Base

UML Class Diagram (module)